$$2x^2 - 15x - 27 = 2x^2 - 18x + 3x - 27$$
$$= 2x(x-9) + 3(x-9)$$
$$= (x-9)(2x+3)$$

Therefore, $(x-9)$ and $(2x+3)$ are the factors of $2x^2 - 15x - 27$

EAVC5

Example 3.31 Factorise $(x+y)^2 + 9(x+y) + 20$

Solution

Let $x + y = p$, we get $p^2 + 9p + 20$

Compare with $ax^2 + bx + c$,

We get $a = 1,\ b = 9,\ c = 20$

product $ac = 1 \times 20 = 20$, sum $b = 9$

Product of numbers $ac = 20$	Sum of numbers $b = 9$	Product of numbers $ac = 20$	Sum of numbers $b = 9$
1×20	21	-1×-20	-21
2×10	12	-2×-10	-12
4×5	9	-4×-5	-9

The required factors are 4 and 5

$\therefore$ we split the middle term as $4p$ and $5p$

$$p^2 + 9p + 20 = p^2 + 4p + 5p + 20$$
$$= p(p+4) + 5(p+4)$$
$$= (p+4)(p+5)$$

Put, $p = x + y$ we get, $(x+y)^2 + 9(x+y) + 20 = (x+y+4)(x+y+5)$

Exercise 3.6

1. Factorise the following:

(i) $x^2 + 10x + 24$ (ii) $z^2 + 4z - 12$ (iii) $p^2 - 6p - 16$

(iv) $t^2 + 72 - 17t$ (v) $y^2 - 16y - 80$ (vi) $a^2 + 10a - 600$

2. Factorise the following:

(i) $2a^2 + 9a + 10$ (ii) $5x^2 - 29xy - 42y^2$ (iii) $9 - 18x + 8x^2$

(iv) $6x^2 + 16xy + 8y^2$ (v) $12x^2 + 36x^2y + 27y^2x^2$

(vi) $(a+b)^2 + 9(a+b) + 18$

3. Factorise the following:

(i) $(p-q)^2 - 6(p-q) - 16$ (ii) $m^2 + 2mn - 24n^2$ (iii) $\sqrt{5}a^2 + 2a - 3\sqrt{5}$

(iv) $a^4 - 3a^2 + 2$ (v) $8m^3 - 2m^2n - 15mn^2$ (vi) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{2}{xy}$

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